Kurtosis and Checking Normality with Q-Q Plots

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Where this fits: Pipeline map step 3 (Understand data). See the Course map.

0 Foundations	1 Frame	2 Get data	**3 Understand**	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Normal distribution.
- **Leads to:** Function transformer; Power transformer; Assumptions of linear regression.
- **Compare with:** Skewness.

1. Overview

Key point: Kurtosis measures how heavy the tails of a distribution are; a Q-Q plot shows whether data follows a given distribution, and it shows heavy or light tails at a glance.

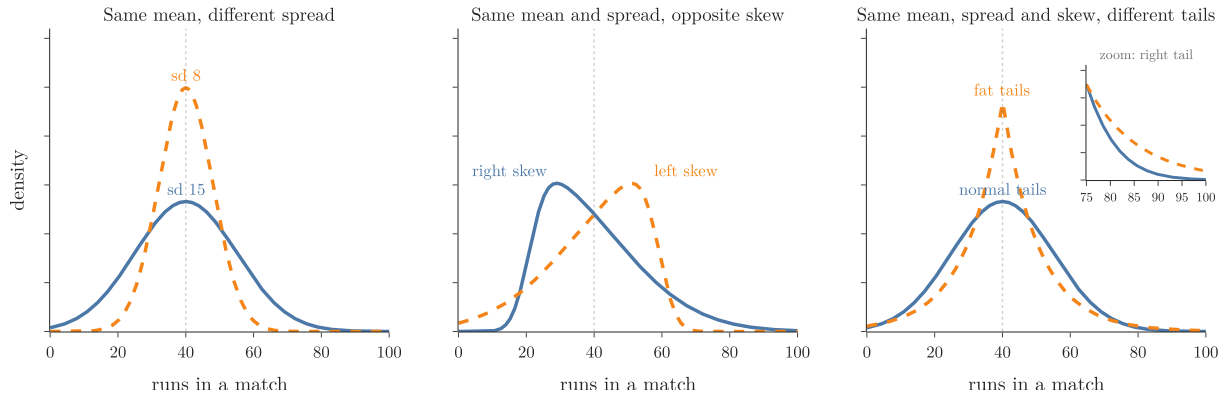


Figure 1: Runs per match in two seasons: each panel keeps the earlier summary numbers equal and changes one more

Figure 1 shows why we need more than one summary number. In each panel the two seasons have the same mean, yet they tell different stories: first the spread differs, then the skew, and finally only the tails.

The first three are known: the mean and the standard deviation (the typical distance from the mean; measures of dispersion) and skewness (how lopsided the curve is; **skewness** (G-1817)). This Note adds the fourth, **kurtosis** (G-1021). The Note's second half answers a common interview question: how do we check whether a **feature** (G-772; one variable of the data, one column of the table) is normal? The main tool is the **Q-Q plot** (G-1596), which also works for distributions other than the normal.

2. Four summary numbers of a distribution

> **Key point:** Mean, standard deviation, skewness and **kurtosis** (G-1021) each add one layer of information about a distribution's shape.

Take a batter who plays 100 matches in each of two seasons and averages 40 runs per match in both. By the mean alone the two seasons are identical. Figure 1 shows three ways in which they can still differ.

1. **Spread (left panel).** In the season with the narrower curve (standard deviation 8), most scores lie close to 40. The batter was more consistent: fewer centuries, but also fewer ducks (scores of 0).
2. **Skew (middle panel).** Same mean, same standard deviation. In one season the batter usually made modest scores with a few big ones (right skew); in the other, usually big scores with a few failures (left skew).
3. **Tails (right panel).** Same mean, same standard deviation, same skew (0). The dashed curve has fatter tails: its density stays above zero further out (see the inset). That season had more extreme scores on both sides, more ducks and more centuries, balanced so that the mean and spread stay the same.

The third difference is what kurtosis measures. Each number is a new perspective on the same data:

Number	Question it answers	Figure 1
Mean	Where is the centre?	the same in all panels
Standard deviation	How spread out are the values?	left panel
Skewness	Is one tail longer than the other?	middle panel
Kurtosis	How heavy are the tails?	right panel

2.1 Statistical moments

> **Key point:** These four numbers are the first four statistical moments of a distribution.

In statistics these numbers are called **moments** (G-1882): averages of the distances from the mean raised to a power. The first four are:

- **1st moment:** the mean.
- **2nd moment:** the **variance** (G-2074), the average squared distance from the mean. The standard deviation is its square root, so the two carry the same information.
- **3rd moment: skewness** (G-1817), from the cubed distances.
- **4th moment: kurtosis** (G-1021), from the distances raised to the fourth power.

There are higher moments, but these four are the ones in common use.

3. Kurtosis: a measure of tailedness

> **Key point: Kurtosis** (G-1021) measures the tailedness of a distribution: how often values far from the mean occur; fat tails mean more outliers.

Kurtosis (from the Greek for "curved, arching") is the fourth **statistical moment** (G-1882). Kurtosis is a measure of the **tailedness** (G-1944) of the probability distribution of a real-valued random variable. Like **skewness** (G-1817), it describes one particular aspect of the distribution's shape.

A **fat tail** (G-757), or heavy tail, is a tail that falls to zero slowly. Far-away values then have a noticeable density, so they occur more often than in a curve whose tails drop quickly. In other words:

- **fat tails:** many values far from the mean, so many outliers;
- **thin tails:** few values far from the mean, so few outliers.

So when someone asks what kurtosis is, the short answer is: how fat the tails of a distribution are, which means how prone it is to outliers.

3.1 Kurtosis is not peakedness

> **Key point: Kurtosis** (G-1021) is about the tails, not the height of the peak; descriptions of it as "peakedness" are wrong.

Many books and websites describe kurtosis as how **peaked** or flat a curve is, or as "peakedness plus tails". Both are incorrect. The statistician Peter Westfall showed that kurtosis has only one clear interpretation: **tail extremity**, meaning either outliers that are present in the data, or a tendency to produce outliers (Westfall 2014).

The reason is in the formula below. Values near the mean contribute almost nothing to it, so the shape of the peak hardly matters; values far from the mean dominate it.

> **Extra:** A fat-tailed curve with the same standard deviation as a normal curve often does have a sharper peak (Figure 3), which is where the "peakedness" idea came from. But we can build distributions with a flat top and huge kurtosis, or a sharp peak and low kurtosis (Westfall 2014). The tails decide, not the peak.

3.2 The kurtosis formula

> **Key point: Kurtosis** (G-1021) is the average of the z-scores raised to the fourth power.

Each value's z-score is its distance from the mean in standard deviations (see **standardization** (G-1874)). Raising z-scores to the fourth power makes far values count enormously. A value 2 standard deviations away contributes

$$2^4 = 16$$

while a value 0.5 away contributes only

$$0.5^4 = 0.0625$$

1. **In words:** standardize every value, raise each z-score to the fourth power, and average.
2. **Formula:** for n values with mean $\bar{x}$ and standard deviation σ (dividing by n), and with m_2 and m_4 the average squared and fourth-power distances from the mean,

$$\text{kurtosis} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{\sigma}\right)^4$$

$$= \frac{m_4}{m_2^2}$$

3. **Example:** two seasons of 8 matches each, both with mean 40 and standard deviation 20.
   - Season A: 20, 20, 20, 20, 60, 60, 60, 60. Every score is exactly 1 standard deviation from the mean, so every $z^4 = 1$:

$$\text{kurtosis of A} = \frac{8 \times 1}{8} = 1$$

   - Season B: six scores of 40, one duck (0) and one 80. Six z-scores are 0 and two are ± 2:

$$\text{kurtosis of B} = \frac{6 \times 0 + 2 \times 2^4}{8}$$

$$= \frac{32}{8} = 4$$

   Same mean, same spread, both symmetric; season B's two extreme scores give it four times the kurtosis.

Figure 2 plays the computation. Watch step 3: raising to the fourth power turns season A's eight z-scores of ± 1 into eight 1s, but turns season B's two z-scores of ± 2 into two 16s. Those two matches alone decide season B's kurtosis; its six scores at the mean add nothing.

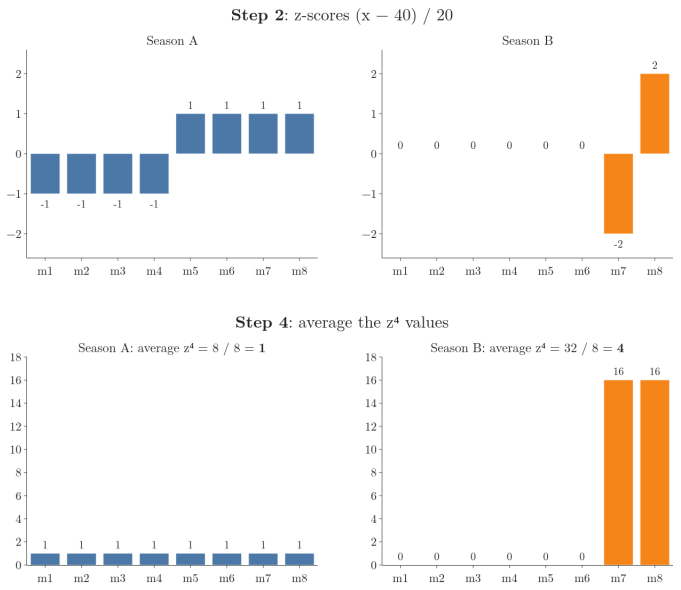


Figure 2: Kurtosis of the two 8-match seasons in steps: the scores, their z-scores, the fourth powers, and the average. Season A: eight 1s, kurtosis 1. Season B: two 16s and six 0s, kurtosis 4.

4. Excess kurtosis and the three types

> **Key point:** Excess kurtosis is **kurtosis** (G-1021) minus 3, the kurtosis of every normal distribution; positive means fatter tails than normal (leptokurtic), negative means thinner (platykurtic), zero means normal-like (mesokurtic).

Every normal distribution, whatever its mean and standard deviation, has kurtosis exactly 3. The mean and standard deviation drop out because kurtosis is built from z-scores, and turning values into z-scores makes every normal distribution the same standard normal curve (see standardizing a normal variable). For that one curve, the average of z^4 is 3 (NIST/SEMATECH e-Handbook §1.3.5.11). So kurtosis is usually reported relative to the normal:

1. **In words:** subtract the normal distribution's kurtosis, 3.
2. **Formula:**
$$\text{excess kurtosis} = \text{kurtosis} - 3$$
3. **Example:**
   - season A has kurtosis 1:
   $$1 - 3 = -2$$
   - season B has kurtosis 4:
   $$4 - 3 = +1$$

Excess kurtosis (G-720) measures how much heavier or lighter a distribution's tails are than those of a normal distribution. Excess kurtosis sorts distributions into three types (Figure 3):

Type	Excess kurtosis	Tails compared with normal	Finance example
Leptokurtic (G-1081; "lepto" = slender)	above 0	fatter: more extreme values, more outliers	a fund with sudden price jumps
Mesokurtic (G-1212; "meso" = middle)	0	the same	a balanced fund
Platykurtic (G-1504; "platy" = broad)	below 0	thinner: fewer extreme values	a stable fund with little price movement

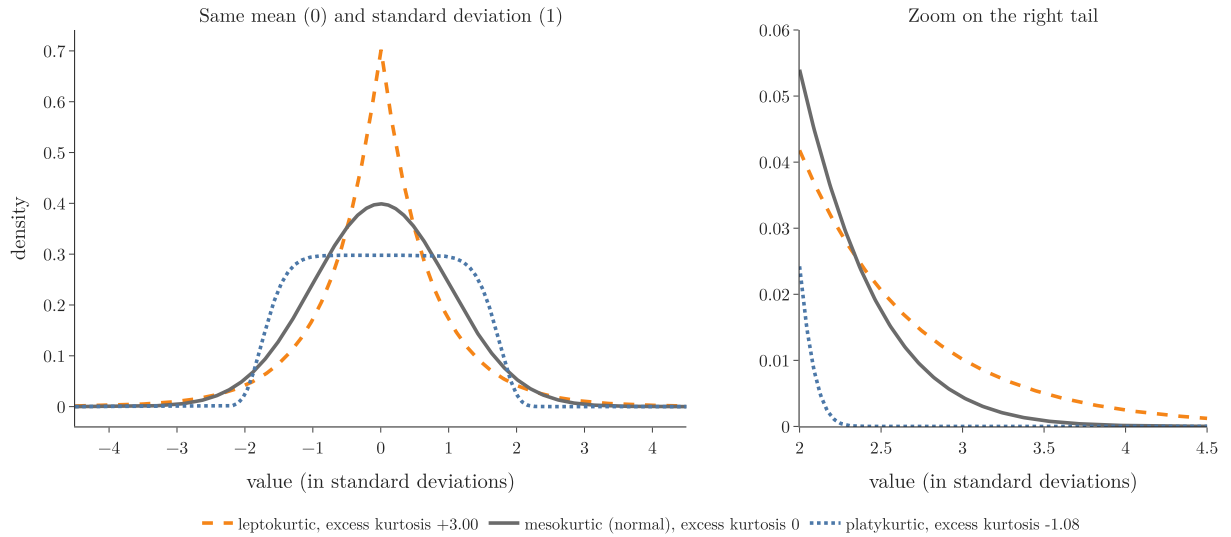


Figure 3: Three distributions with mean 0 and standard deviation 1 that differ only in their tails

In Figure 3 the three curves have the same mean, the same standard deviation and no skew. The zoom on the right shows what differs. Beyond 3 standard deviations the leptokurtic curve has 1.44% of its area, the normal curve 0.27% and the platykurtic one practically none.

The prime example of a mesokurtic distribution is the normal distribution itself, for any values of its parameters. The uniform distribution (next Note) is strongly platykurtic, with excess kurtosis -1.2: it has no tails at all.

> **Python:** Kurtosis in scipy and pandas.
```
import pandas as pd
from scipy import stats

b = [40, 40, 40, 40, 40, 40, 0, 80]
```

```
# excess kurtosis by default (fisher=True)
stats.kurtosis(b)                  # 1.0
# plain kurtosis
stats.kurtosis(b, fisher=False)    # 4.0
# pandas: excess, with a small-sample correction
pd.Series(b).kurt()                # 3.5
```

scipy (by default) and pandas report **excess** kurtosis and simply call it "kurtosis", so a normal feature shows about 0, not 3. pandas also corrects for sample size, which matters a lot with 8 values (3.5 against 1.0) and little with hundreds (0.478 against 0.461 for 500 normal values in the Notebook).

5. Where kurtosis matters: kurtosis risk

Key point: In finance, high **kurtosis** (G-1021) in an asset's returns means a higher chance of extreme gains and extreme losses, called kurtosis risk.

Kurtosis is not needed in every analysis, but finance is one field that relies on it. **Kurtosis risk** (G-1020) is the risk that comes from the possibility of extreme outcomes, the **fat tails** (G-757), in the distribution of returns of an asset or a portfolio.

Picture the distribution of returns of a mutual fund. A fat tail on both sides means many investors made a lot of money and many lost a lot. Such a fund is called **volatile** (G-2094).

So analysts plot the return distribution of an asset and compute its kurtosis. If the kurtosis is high, investors are warned that extreme gains and extreme losses are both more likely than a normal curve would suggest, and they can adjust their strategy for it.

6. Is a feature normally distributed?

Key point: Besides plots, a statistical test such as Shapiro-Wilk checks normality with a p-value; it is best read together with a **Q-Q plot** (G-1596).

Many methods assume that a feature is normally distributed, so "how do we know whether a feature is normal?" is a common interview question. The visual checks, a density plot and a Q-Q plot, are taught in checking whether a column is normal; the Q-Q plot is the most informative and gets a second look below.

The third way is a **statistical test** (G-1881). The **Shapiro-Wilk test** (G-1788) (see the Shapiro-Wilk test) and the **Anderson-Darling test** (G-200) decide with the help of a **p-value** (G-1433; the chance of data at least this far from normal if the data really were normal); they come after hypothesis testing.

Extra: For the 150 iris sepal lengths of Section 7.3, `stats.shapiro(sepal)` gives a p-value of 0.010. At the usual 5% level the test rejects normality, even though the histogram looks roughly like a bell. With large samples these tests flag even small departures from normality (Ghasemi and Zahediasl 2012), so they are best read together

with a Q-Q plot.

A gotcha follows from how these tests work: with large samples they flag even small departures from normality (Ghasemi and Zahediasl 2012). Figure 4 tests that claim on data of one fixed shape: a Student t distribution with 10 degrees of freedom (a setting that controls the tails: fewer degrees of freedom give fatter tails), a bell with tails only slightly fatter than normal. For each sample size we draw 200 samples and count how often Shapiro-Wilk rejects normality at 5 percent:

- with 20 values it rejects 12 percent of the samples, close to the 5 percent it would reject for truly normal data;
- with 500 values, 66 percent;
- with 2,000 or more, every sample.

The shape never changed; only the sample grew. A small p-value on a large sample can mean a departure too small to matter, which is why we also look at the Q-Q plot.

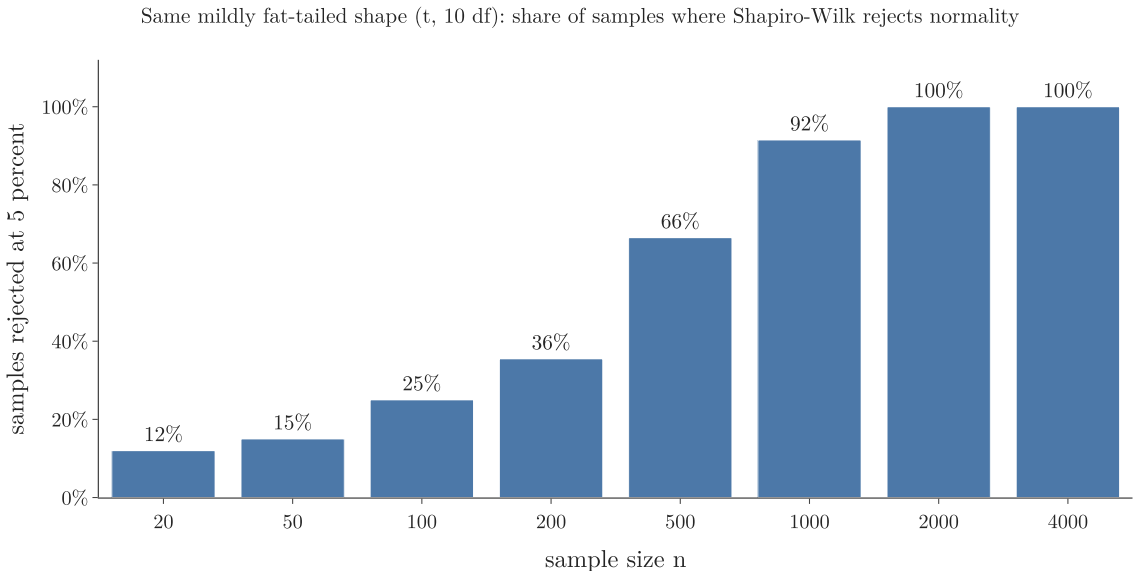


Figure 4: Share of 200 samples in which the Shapiro-Wilk test rejects normality at 5 percent, for samples of 20 to 4,000 values from the same mildly fat-tailed t distribution.

7. Building a Q-Q plot

Key point: A **Q-Q plot** (G-1596) pairs each sorted data value with the matching quantile of a theoretical distribution; the quantiles can come from cutting the curve into equal-area strips, or from the percentiles of a large generated sample.

How to read a Q-Q plot, and a five-value build, are in how a Q-Q plot is built and reading a Q-Q plot. Here we build the plot twice: first on 15 values, where every point can be followed by eye, and then from percentiles (percentiles: the value below which a given share of the

data lies), which needs no formula.

7.1 Fifteen values, one point each

> **Key point:** Cut a normal curve into strips of equal area; the cuts are the normal quantiles, and each one is paired with one sorted data value.

Take 15 sepal lengths: every tenth one of the 150 sorted iris values (Section 7.3), from 4.5 cm to 7.6 cm. Is their shape normal? Figure 5 answers in four steps.

1. **Sort the data.** Each of the 15 sorted values is one **quantile** (G-1599) of the data: the smallest, the second smallest, and so on.
2. **Cut a normal curve into equal-area strips.** Any normal curve will do; we take mean 0 and standard deviation 1. Fifteen cuts make 16 strips, and each strip holds the same share of the area, 1/16, so a normal value is equally likely to land in any strip. The strips at the edges are wide, because the curve is low there and a strip needs more width to collect its 1/16. The strips in the middle are narrow, because the curve is high. The positions of the cuts are the **theoretical quantiles** (G-1968): -1.53, -1.15, -0.89, ..., 1.53. Cut number k has the share $k/16$ of the area on its left. For the first cut:

   $$1/16 = 0.0625$$

   The **z-table** (G-2142) (a table of areas under the standard normal curve) gives:

   $$\Phi(-1.53) = 0.063$$

   where $\Phi(z)$ is the area under the standard normal curve to the left of z. So the first cut sits at -1.53.
3. **Pair them, one point each.** The smallest value, 4.5 cm, goes with the first cut, -1.53: a horizontal dotted line from 4.5 and a vertical dotted line from -1.53 cross at the first point. The second value, 4.8 cm, goes with -1.15; the third, 5.0 cm, with -0.89; and so on for all 15.
4. **Draw a straight line through the points.** If the data is normal, its values are spaced like the cuts: crowded in the middle and spread out at the ends. The points then fall on a straight line.

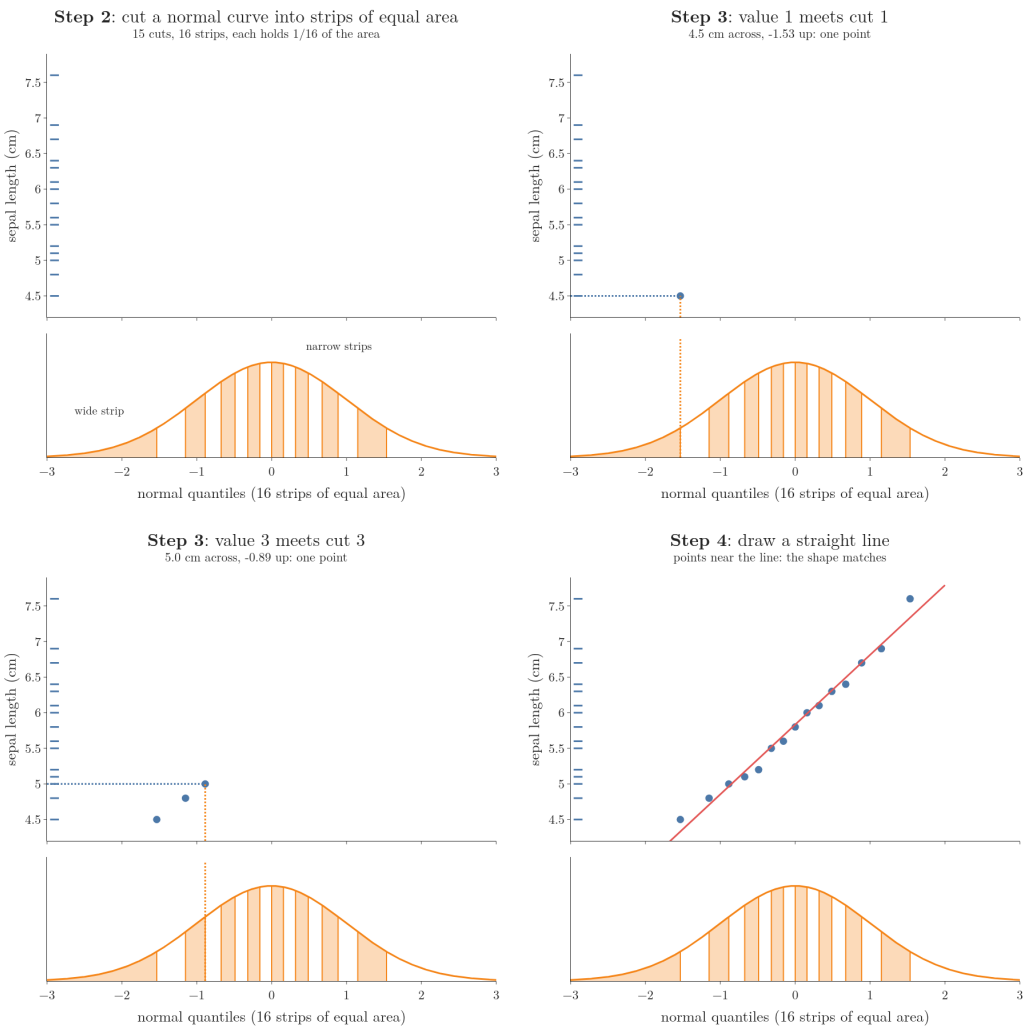


Figure 5: A Q-Q plot from 15 sepal lengths. A normal curve is cut into 16 strips of equal area; each cut (vertical dotted line) meets one sorted value (horizontal dotted line) at a point; a straight line is drawn through the 15 points. Idea after StatQuest, "Quantile-Quantile Plots (QQ plots), Clearly Explained!!!".

In Figure 5, watch step 2: the two outer strips are several times wider than the middle ones. In the last frame the 15 points stay close to the line, so these values are roughly normal.

With 150 or more values we do the same pairing with percentiles, as follows.

7.2 The steps with percentiles

> **Key point:** Our data on the y axis, a theoretical distribution on the x axis, one point per **quantile** (G-1599).

We compare our data X with a **theoretical distribution** Y: one whose type we already know, here the normal distribution (see the normal distribution). Figure 6 shows the steps.

1. **Theoretical data.** Generate many values from the theoretical distribution, say 1,000 values from a normal distribution with mean 0 and standard deviation 1.
2. **Sort both and take quantiles.** Sort our data and compute its percentiles: 1st, 2nd, . . . , 99th. Do the same for the theoretical data.
3. **Pair them.** The 1st percentile of Y with the 1st percentile of X, the 2nd with the 2nd, and so on.
4. **Plot the pairs. Theoretical quantile** (G-1968) on the x axis, data quantile on the y axis.

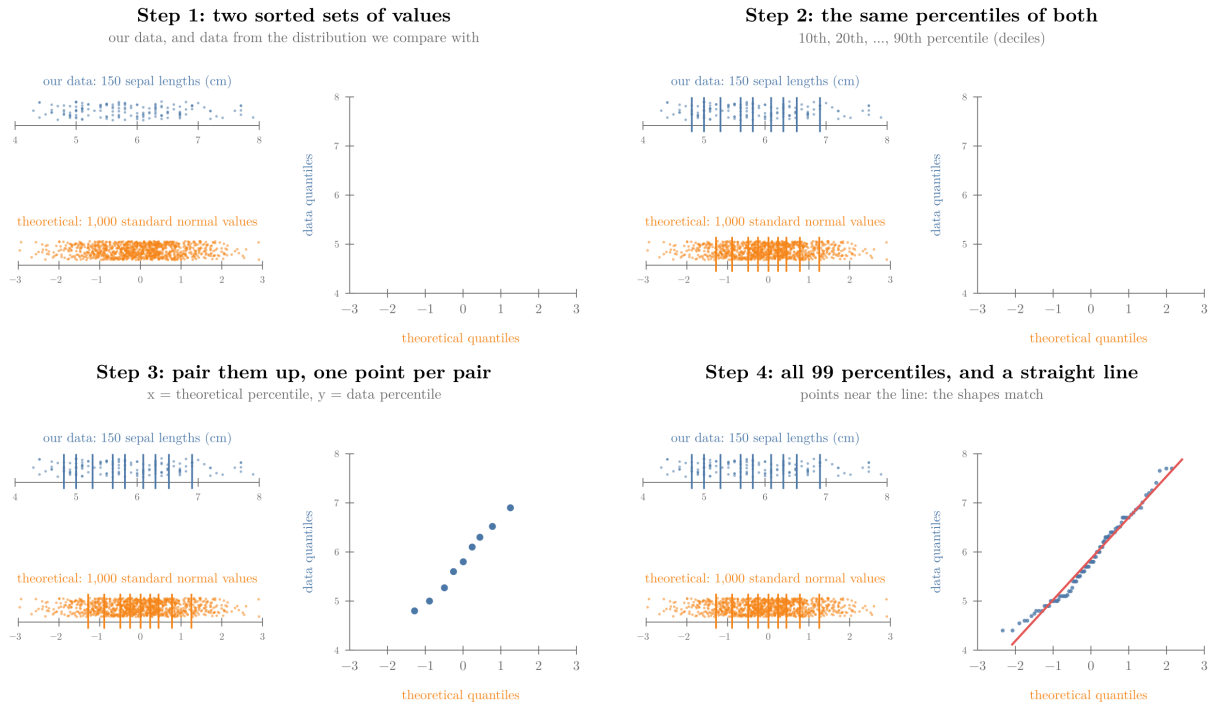


Figure 6: Building a Q-Q plot: sort both sets, take the same percentiles, plot them in pairs

Points on one straight line still mean that X has the same shape as Y. The line does not need to be the diagonal $y = x$. Our data can have any mean and standard deviation; those only move and tilt the line (Section 7.4).

7.3 The iris sepal lengths

> **Key point:** The 150 sepal lengths lie close to the line in the middle and stray at the ends: roughly normal, but not perfectly.

The iris dataset holds measurements of 150 flowers. Its `sepal length` feature looks like a bell in a density plot. Figure 7 (left) builds its **Q-Q plot** (G-1596) by hand, with 99 percentiles of the data against 99 percentiles of 1,000 standard normal values. The right panel is the ready-made version.

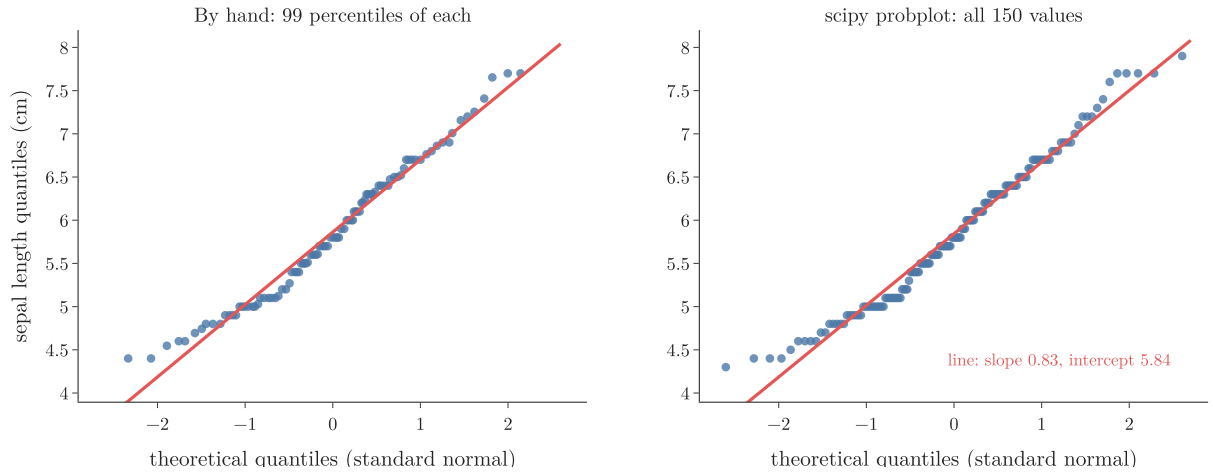


Figure 7: Q-Q plot of the iris sepal lengths against the normal distribution, by hand and with scipy

The points follow the line closely in the middle, but not every point touches it, and the lowest and highest values drift away. So the sepal lengths are roughly normal, not perfectly normal. The steps of points come from the data being rounded to 0.1 cm: the 150 sepal lengths take only 35 distinct values, so many neighbouring percentiles share the same value and line up horizontally.

Python: A Q-Q plot by hand, then the shortcut.

```
import numpy as np
from scipy import stats
from sklearn.datasets import load_iris

sepal = load_iris().data[:, 0]
rng = np.random.default_rng(42)
theo = rng.normal(0, 1, 1000)
pct = np.arange(1, 100)        # 1st to 99th
y_quant = np.percentile(sepal, pct)
x_quant = np.percentile(theo, pct)
# plot x_quant against y_quant as a scatter

# shortcut: sorting and quantiles done for us
(osm, osr), (slope, icpt, r) = \
    stats.probplot(sepal, dist="norm")
```

`probplot` uses exact normal **quantiles** (G-1599) instead of a generated sample, and one point per data value. The Notebook (`MA-028-kurtosis-and-qq-plots.ipynb`) draws both panels with Plotly.

7.4 Which reference line?

> **Key point:** The diagonal $y = x$ only fits standardized data; for raw data use a line fitted through the points.

statsmodels has a one-line **Q-Q plot** (G-1596), `sm.qqplot(x, line=...)`, with four choices of reference line:

- `"45"`: the diagonal $y = x$;
- `"s"`: a standardized line, with the data's standard deviation as slope and its mean as intercept;
- `"r"`: a regression line fitted through the points;
- `"q"`: a line through the first and third quartiles.

The diagonal only works when the data has mean 0 and standard deviation 1, or when we pass `fit=True`, which standardizes the data first (statsmodels `ProbPlot` docs). The sepal lengths have mean 5.84 and standard deviation 0.83, so their points lie around the line with slope 0.83 and intercept 5.84 (Figure 7, right), far from $y = x$. For raw data we use `"s"`, `"r"` or `"q"`; scipy's `probplot` always draws the regression line.

> **Python:** `sm.qqplot` draws with matplotlib, which these Notes do not use. The same numbers come from `sm.ProbPlot(x, fit=True)`, whose `theoretical_quantiles` and `sample_quantiles` can be plotted with Plotly.

8. Reading tails on a Q-Q plot

> **Key point: Fat tails** (G-757) make the points leave the line at both ends, outwards; thin tails bend them back in, in an S shape.

The basic shapes, including fat tails leaving the line outwards at both ends, are in reading a Q-Q plot (Figure 5). **Kurtosis** (G-1021) explains that fat-tail shape: a leptokurtic feature has more extreme values on both sides than a normal one. Such a curve often also looks too peaked in the middle, but it is the tails that move the points (Section 3.1).

Figure 8 shows the fat-tail shape on 1,000 standardized values from the peaked, fat-tailed **Laplace distribution** (G-1044), the Notebook's check. The middle point sits on the line, while the lowest and highest values are -5.41 and 4.82 where the normal **quantiles** (G-1599) are only -3.20 and 3.20: both ends leave the line outwards.

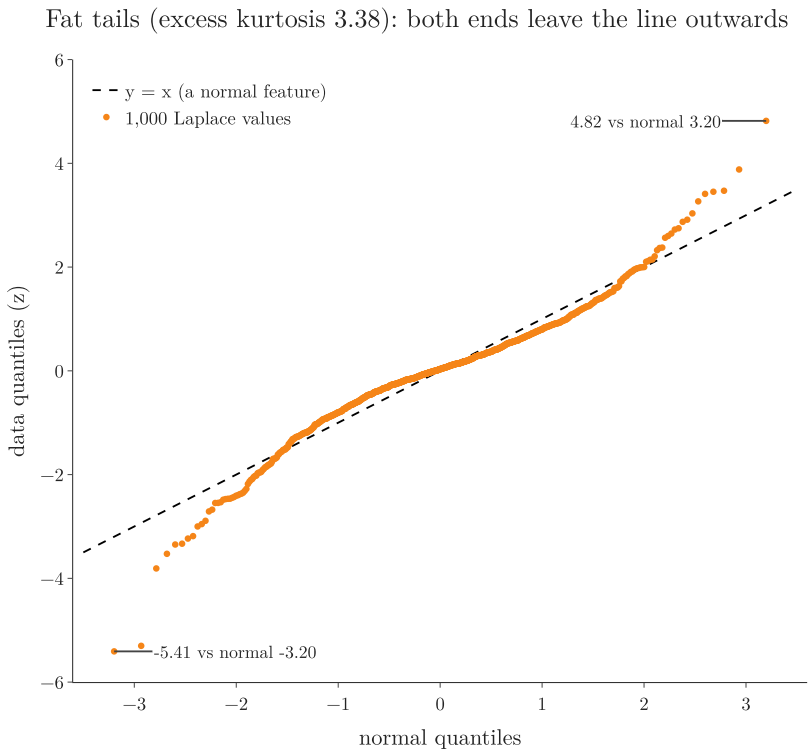


Figure 8: Q-Q plot of 1,000 standardized Laplace values against the normal. The middle follows the dashed line; both ends bend away from it, outwards.

Thin tails (platykurtic) give the opposite shape, which Figure 9 (left) shows for uniform data:

- the middle points sit near the line;
- the end points flatten out, inside the line, in an S shape: the extreme values are less extreme than normal;
- with mild thin tails the points stay near the line and stray only slightly.

The more the points leave the line, the further the data is from normal.

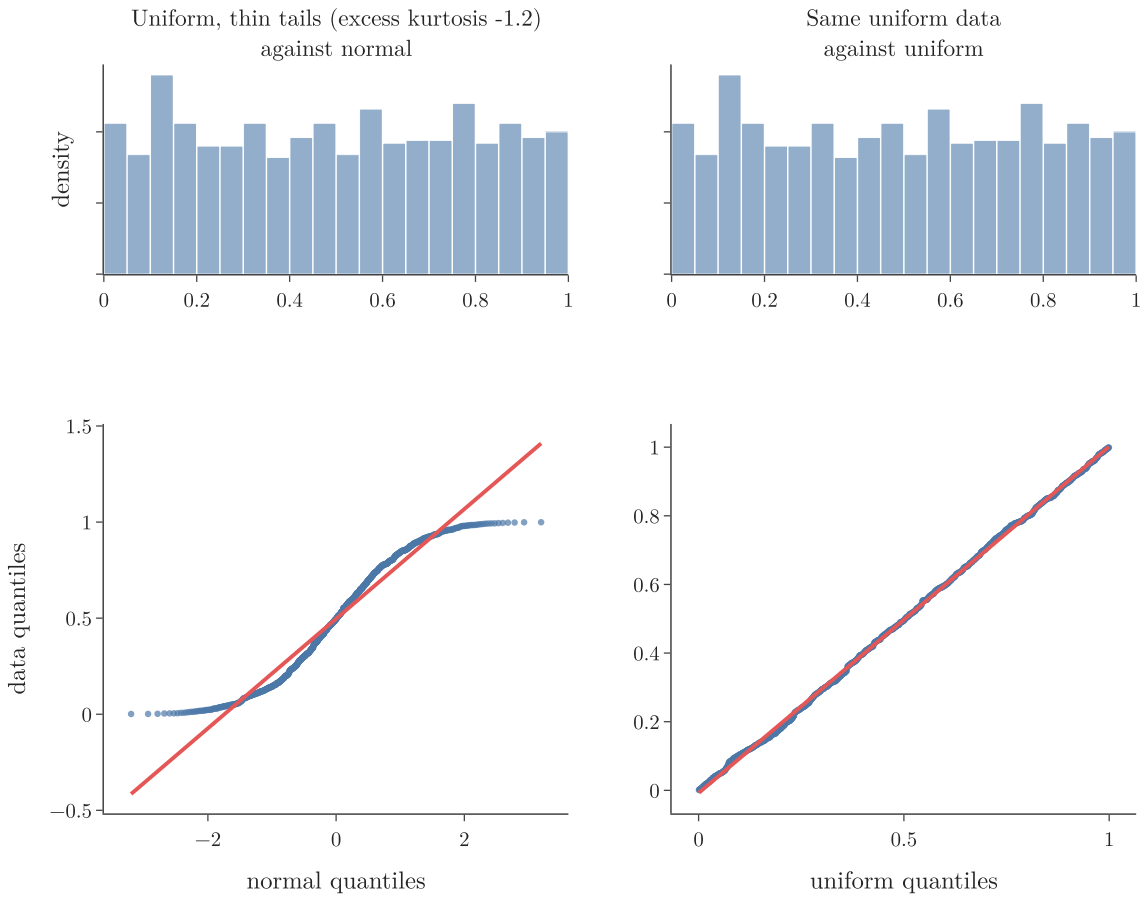


Figure 9: Top: histograms; bottom: Q-Q plots. Uniform data against the normal, then the same data against the uniform

9. Q-Q plots for other distributions

> **Key point:** A **Q-Q plot** (G-1596) can compare data with any theoretical distribution, not only the normal: uniform, log-normal, Pareto and others.

A common misunderstanding is that Q-Q plots can only detect normal distributions. By definition a Q-Q plot compares two distributions, so the theoretical one can be anything. Only the **quantiles** (G-1599) on the x axis change.

Figure 9 shows 1,000 values drawn from a **uniform distribution** between 0 and 1, where every value in the range is equally likely (the uniform distribution). The histogram looks flat.

- **Against the normal (left):** an S-shaped curve, the thin-tail shape. The data is clearly not normal.
- **Against the uniform (right):** almost every point is on the line. The data is uniform.

Python: Pass the theoretical distribution with `dist`.

```
uni = rng.uniform(0, 1, 1000)
(osm, osr), _ = stats.probplot(uni, dist=stats.uniform)
# statsmodels: sm.ProbPlot(uni, dist=stats.uniform)
```

The default `dist` is the normal distribution, which is why it was not passed in Section 7.

The same idea checks for the log-normal and Pareto distributions.

10. Summary

Idea	Meaning	Example	Why it matters
Moments	Mean, variance, skewness, kurtosis	1st to 4th	each adds one layer: centre, spread, lopsidedness, tails
Kurtosis	Average of z^4: tail heaviness	season A 1, season B 4	the fourth power makes far values dominate, so two extreme matches decide B's value
Excess kurtosis	Kurtosis -3; normal $= 0$	A -2, B $+1$	every normal distribution has kurtosis 3, so the sign says fatter or thinner tails than normal
Leptokurtic	Excess above 0: fat tails, more outliers	volatile fund	extreme gains and losses are more likely than a normal curve suggests (kurtosis risk)
Platykurtic	Excess below 0: thin tails	uniform, -1.2	extreme values are rarer than normal
Q-Q plot	Data quantiles against theoretical quantiles	iris sepal length vs normal	shows at a glance whether data follows a distribution, and where it departs

- **Kurtosis** (G-1021) is about tails, not peakedness, because values near the mean add almost nothing to the average of z^4.
- Most software reports **excess kurtosis** (G-720); a normal feature gives about 0, so do not expect 3 from scipy or pandas.
- Three ways to check normality: plot, **Q-Q plot** (G-1596), statistical test, so read the test together with a Q-Q plot, because with large samples the test flags even small departures.
- Points on a straight line: same shape. The diagonal $y = x$ only fits standardized data, so for raw data use a fitted line (`"s"`, `"r"` or `"q"`).
- Fat tails leave the line outwards at both ends; thin tails bend inwards (S shape), because the extreme values are more, or less, extreme than the normal **quantiles** (G-1599).
- Any theoretical distribution can go on the x axis of a Q-Q plot, so the same tool checks for uniform, log-normal or Pareto data.
- This answers the opening question: kurtosis measures how heavy the tails are, and a Q-Q plot shows whether data follows a given distribution and whether its tails are

heavy or light.

11. Sources

Built from

- CampusX, "Session 42 - Non-Gaussian Probability Distributions | DSMP 2023", YouTube, https://www.youtube.com/watch?v=U6QCc_3zgUk
- StatQuest with Josh Starmer, "Quantile-Quantile Plots (QQ plots), Clearly Explained!!!", YouTube, https://www.youtube.com/watch?v=okjYjClSjOg

Other references

- Ghasemi, A. and Zahediasl, S. (2012). "Normality Tests for Statistical Analysis: A Guide for Non-Statisticians." *International Journal of Endocrinology and Metabolism* 10(2).
- statsmodels documentation, `ProbPlot` (`fit=True`) and `qqplot` (`line` options).
- NIST/SEMATECH *e-Handbook of Statistical Methods*, §1.3.5.11, "Measures of Skewness and Kurtosis" (the kurtosis of a standard normal distribution is three). https://www.itl.nist.gov/div898/handbook/eda/section3/eda35b.htm
- Westfall, P. H. (2014). "Kurtosis as Peakedness, 1905-2014. R.I.P." *The American Statistician* 68(3).

12. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Kurtosis (G-1021)	How heavy the tails of a distribution are compared with a normal curve; the fourth moment.
Statistical moments (G-1882)	Averages of distances from the mean raised to a power: mean, variance, skewness, kurtosis.
Tailedness (G-1944)	How much probability lies far from the mean, in the tails.
Fat tail (heavy tail) (G-757)	A tail that falls to zero slowly, so extreme values are relatively common.
Excess kurtosis (G-720)	Kurtosis minus 3, so a normal distribution scores 0.
Leptokurtic (G-1081)	Said of a distribution whose tails are fatter than the normal's (excess kurtosis above 0), so extreme values and outliers are more common.

Term	Meaning
Mesokurtic (G-1212)	Said of a distribution whose tails are like the normal's (excess kurtosis 0), so extreme values are about as common as under a normal distribution.
Platykurtic (G-1504)	A distribution with thinner tails than the normal, so fewer extreme values; its excess kurtosis is below 0.
Kurtosis risk (G-1020)	In finance, the risk of extreme gains or losses from fat-tailed returns.
Volatile (G-2094)	Changing quickly and unpredictably.
Anderson-Darling test (G-200)	A statistical test of whether data follows a given distribution, such as the normal; like the Shapiro-Wilk test, it decides with a p-value.
Theoretical distribution (G-1967)	The known distribution that data is compared with, for example on a Q-Q plot.
Skewness (G-1817)	A number for how lopsided a distribution is: 0 symmetric, positive right tail, negative left tail.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Q-Q plot (G-1596)	A plot of a column's sorted values against the values a theoretical distribution, often the normal, would have; points on the line mean the data follows it.
Variance (of data) (G-2074)	The average squared distance of the values from their mean; the square of the standard deviation; divide by n for a population (NumPy default) or $n-1$ for a sample (pandas default).
Standardization (G-1874)	Scaling a column to mean 0 and standard deviation 1 (subtract the mean, divide by the standard deviation), so features measured on different scales become comparable.
Statistical hypothesis test (G-1881)	A method that uses sample data to decide whether a claim about a whole population (a hypothesis about a population parameter) is supported strongly enough; it is a form of statistical inference.

Term	Meaning
Shapiro-Wilk test (G-1788)	A statistical test of whether data follows a normal distribution: it takes the values and returns a statistic and a p-value, and a small p-value says the data is unlikely to be normal.
P-value (G-1433)	The probability, assuming H_0 is true, of getting a sample as or more extreme than ours.
Quantiles (G-1599)	Values that cut sorted data into equal-sized groups; a fraction p of the values lie below the quantile $Q(p)$.
Theoretical quantile (G-1968)	Where a value would sit if the data were perfectly normal (the horizontal axis of a Q-Q plot).
Z-table (G-2142)	A table that gives, for many values of z, the probability that a standard normal value falls below z, written $\Phi(z)$; used to find normal probabilities.
Laplace distribution (G-1044)	A peaked, heavy-tailed distribution with density $e^{-\lvert x-\mu\rvert/b}/(2b)$.